Soak Your PCB: a Design Activity for Hands-On Learning of the Electrochemical Interface Impedance

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***Abstract*—A newly introduced laboratory experience addressing the practical understanding of the impedance of the non-Faradaic electrochemical interface, within a bioelectronics class, is presented. The project is carried out by teams of M.Sc. students (3 to 5 persons) of mixed backgrounds (electronics and biomedical engineering). It comprises two phases. First, a design phase, in which a PCB hosting planar gold coated electrodes, to be sized in order to properly measure the ionic resistance of the solution (below 10 MHz), and a low-noise transimpedance stage to interface to a USB network analyzer are designed. Then, during the experimental activity impedance spectroscopy is performed with the electrodes in saline solution, coupled to microfluidics, and data are analyzed to estimate the solution conductivity and compared with simulations.**

Keywords—impedance spectroscopy, educational laboratory.

I. INTRODUCTION

A. *Motivation*

The enhancement of traditional Electronics Engineering (EE) courses with hands-on laboratory activities belongs to a well consolidated trend [1-4]. Students can complement the acquisition of theoretical knowledge with practical experience. Experimental activities, often performed in small teams, are perceived as more engaging than passive attendance to classroom lectures. While the fundamentals of the discipline are usually metabolized during traditional blackboard lessons (with mathematical formalism) and quantitative paper-and-pencil exercises, the use of software tools and implementation details are better conveyed during practical activities. The majority of topics of EE, in particular of circuits and systems curricula (signals, sensors, analog circuit analysis and design, digital circuit and programmable devices such as microcontroller and FPGAs) can be easily translated into a wide range of laboratory experiences. Furthermore, laboratory sessions offer important additional skills including the capability to work in teams, speak and present in public, predict and analyze experimental results and get acquainted with instrumentations (power supply, signal generators, oscilloscopes, spectrum and network analyzers).

This paradigm can be pushed even further, reaching a stage where a traditional course is transformed into a sequence of hands-on lessons aiming at the development of specific group project. There are several evidences of the advantages of project-based learning (PBL) approaches [5,6]. However, when the time allocated to the laboratory is limited, students work on commercial boards, mostly configuring or programming them [7,8] with no PCB or analog design.

In this paper we focus on a bioelectronics course at master level. The course presents biosensors (in particular those based on molecular affinity), microfabrication and microfluidics, optical and electrochemical ligand detection and the challenges of low-noise (sub pA with kHz bandwidth) current readout.

B. *Target Students and Learning Goals*

The 50-hour course is attended by master students belonging to different curricula: 50% of EE, 40% of biomedical engineering and 10% of other engineering programs (physics and nanotechnology/materials science). Given the different backgrounds, a fully PBL approach is not feasible in this case. In fact, a common theoretical baseline has to be learnt by means of frontal teaching.

The expected learning outcomes of this elective activity are the familiarization with the PCB design flow, instrumentation and experimental acquisition of impedance spectra in liquid.

II. THEORETICAL BACKGROUND

The main concepts whose interiorization is promoted by this activity are related to the properties of the metal/electrolyte interface, in particular of its equivalent impedance. Understanding the behavior of the interface between an electrode and an electrolytic solution (commonly a water solution with ions) is pivotal in several applications including:

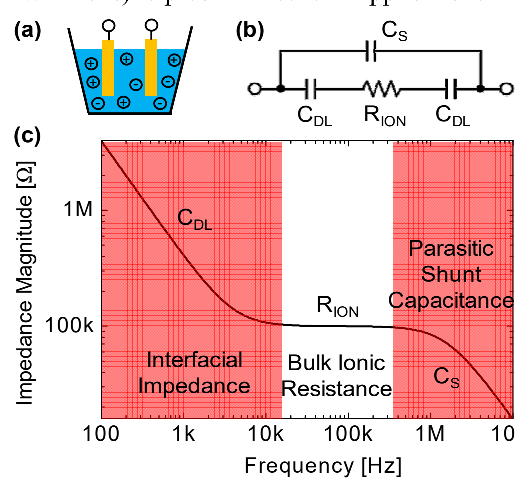

Fig. 1. Electrochemical interface between metal electrodes and ionic solution (a): small-signal equivalent model (b) and ideal impedance spectrum (c).

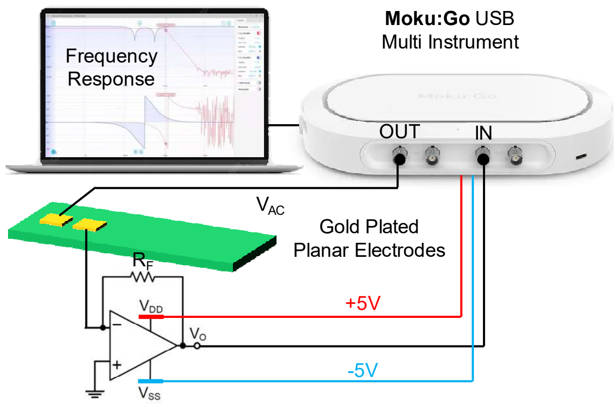

Fig. 2. Experimental setup for performing electrochemical impedance spectroscopy of planar gold electrodes on a PCB hosting the TIA stage.

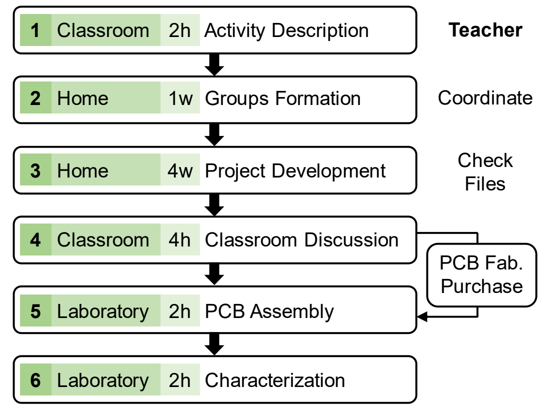

Fig. 3. Sequence of the lab activity steps performed in the classroom, in laboratory (with no need for additional facilities) and at home in teams.

(i) water quality monitoring (salinity) [9], (ii) modeling batteries [10-13], and (iii) biochemical sensing, applied for instance to detection of biological cells [14,15].

The impedance between two electrodes soaked into an electrolytic solution (Fig. 1a) is composed of both bulk and interfacial components [16]. The bulk of an ionic conductor is described by its resistance R_{ION} which depends on the amount and mobility of the different ionic species in solution. The presence of an electrode, biased at a given potential, in solution changes the distribution of ions in its proximity. Ions are attracted towards the electrode surface by the Coulomb force and a multilayer structure is formed. Ions in the first layer are adsorbed on the surface and cannot move (inner Helmholtz plane, 0.1 nm), while the second and third layer (outer Helmholtz plane, 0.2 nm) have some mobility. The density of ions decreases along the diffuse layer with an exponential function of the distance towards the bulk, described by Debye characteristic length. Each of this charge layers can be sees as a plate of a capacitor. Capacitors are in series forming the double layer capacitance (C_{DL}). This is a nonlinear capacitor, since its value depends on the DC bias applied across the interface. Thus, for a given bias point and in absence of redox reactions at the interface, the equivalent impedance model (Fig. 1b) is thus composed of the series of C_{DL} (often a fractional pseudo capacitance Constant Phase Element, CPE) and R_{ION} [17].

III. Laboratory Activity

A. Circuit Design Goals

The goal of the team activity is to design a set of planar electrodes and to measure their impedance spectrum when soaked into a water-based saline solution, in particular to measure R_{ION}. In order to avoid spurious redox reactions, the stimulation voltage V_{AC} has to be smaller than 100 mV and the electrodes have to be made of a noble metal, in particular gold plated. Despite the higher cost, the standard electrochemical gold finish for the PCB (ENIG, electroless nickel immersion gold) is the best option.

As pictured in Fig. 2, the instrument used for the characterization of the impedance is the Moku:Go by Liquid Instruments. It is an USB-controlled, FPGA-based device with two input and output channels (20 MHz analog bandwidth, sampling at 125 MSa/s) conceived for educational use. It can be configured in a variety of virtual instruments: waveform generator, data logger, oscilloscope, spectrum analyzer and frequency response analyzer (FRA). An optional lock-in mode is also present. However, since this was not available at the time of the laboratory, we used the FRA mode to measure the impedance spectrum. Since the instrument input is a voltage signal, an external input stage is necessary. Thus, the task of the students is to design on the same board the electrodes and the transimpedance (TIA) stage. One electrode is driven by the sinusoidal stimulation V_{AC} directly generated at the instrument output, while the other one is connected to the TIA input to convert the input current into a voltage fed to the instrument input. The instrument (in variant M1) also includes a dual programmable power supply able to generate ±5 V with 150 mA current capability. This is very convenient to supply the TIA circuit with no need for external units.

A range of values for the water conductivity is given to the students. The minim conductivity is ideally 0 if ultrapure deionized water is taken. The maximum conductivity is limited by the solubility of salt in water (for instance of 40 g in 100 cm^3 of water for NaCl at room temperature). We take here 5 S/m as maximum conductivity (0.5 □ 0.8 S/m for drinking water, 1.5 S/m for physiological buffer solution, 3 □ 5 S/m for seawater). Thus, for a given value of water conductivity σ, the area and the distance of the electrodes determine their impedance spectrum. The value of R_{ION} can be estimated by means of *conformal mapping*. These are mathematical transformations that map the electric field from the coplanar case to the parallel-plate one, where the field is uniform. They require numerical solutions of elliptic integrals. Students are provided with a paper [18] describing the equations and with a Matlab code running an example for interdigitated electrodes (both in the case semi-infinite and finite layer on the electrodes) that can be adapted to their geometry. The double layer capacitance C_{DL} can be estimated by multiplying the electrode area by the specific capacitance (~0.1 pF/μm^2 for gold and σ = 1.5 S/m and scaling with the square root of σ).

The challenge is to measure R_{ION} within the instrument bandwidth. Hence students have to properly size electrode geometry to place the resistive plateau (defined at low frequency by C_{DL} and at high frequency by some parasitic

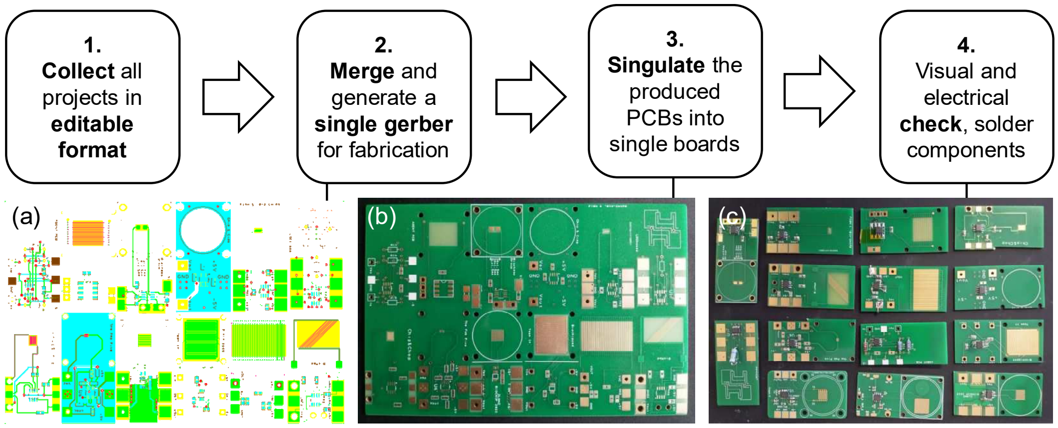

Fig. 4. Global PCB realization process flow. It is similar to MultiProject Wafer (MPW) shuttle runs.

parallel capacitance ~pF) in the range between a few Hz and ~10 MHz. Then, the main design parameter is the TIA feedback resistor R_F. It sets all the key performance: the conversion gain (R_F), bandwidth ($1/2\pi \cdot R_F \cdot C_{stray}$) and noise ($4kT/R_F$) [19]. It is in the ~kΩ range and has to be carefully chosen considering all the constraints: maximization of the signal within the instrument input swing ($|V_{AC} \cdot R_F/R_{ION}| = 5V$), minimization of noise, stability and achievement of a bandwidth compatible with the spectral region to be explored.

B. Activity Plan

The activity articulates in several steps (Fig. 3). The first step is the presentation of the activity in the classroom to the students (2 hours). Design goals and constraints are illustrated during a two-hour lesson. Brief indications about the software tools are provided along with guideline for simulations and PCB design. Students are invited to group into teams with at least one EE and one biomedical student to promote multidisciplinarity. After one week, the teacher checks and trims the composition of the teams. Then, the design activity at home in teams starts. Teams are given a development time between 4 and 6 weeks. By this deadline, teams submit to the teacher the design files. Each group prepares a 5-minute presentation of its design which is then discussed by the teacher and other teams in the classroom. After this important shared discussion (4 hours, crucial check to avoid severe implementation mistakes), the teacher merges the PCB layouts into a single board (Fig. 4) and submit it for fabrication. The manufacturing of a single large PCB is much less expensive than individual small PCBs of the same total area. When the master PCB is received, it is cut into individual boards assembled by the teams during the first day in the laboratory (2 hours) and checked with a multimeter. On the second and final laboratory day (2 hours) the PCB is tested with an oscilloscope (first with a test resistor) and then soaked into a saline solution of known conductivity, whose impedance spectrum is measured with the FRA and compared with the one expected from simulations.

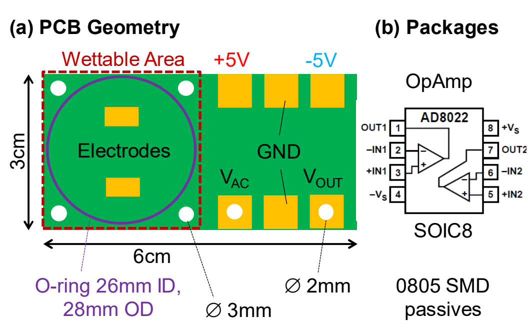

Fig. 5. Geometrical contraints for the PCB hardware design.

Some tasks are mandatory (electrodes and R_F sizing, conformal mapping estimation, PCB design, assembly and characterization), while others are optional such as: (i) design of a microfluidic system to be coupled to the PCB to wet the electrodes in a controlled way, (ii) finite-element simulations of the impedance of the electrodes, (iii) fitting of the measured spectra to extract the lumped equivalent parameters.

C. Design Constraints

During the introductory lecture students are provided with the following constraints concerning: (*i*) water conductivity (σ 0 □ 5 S/m), (*ii*) instrument properties (input voltage range, bandwidth, supply voltage), (*iii*) the PCB. The latter hardware constraints are shown in Fig. 5. They include the PCB layout and the components. Since all 2-layer PCBs will be merged into a single file, they must have the same size (3 cm × 6 cm to fit inside a beaker). Electrodes must be placed in the wettable area (half of the PCB), while electronic components will be placed in the dry half. An area of 3 cm × 3 cm is enough to place a SOIC-8 opamp, passive components and pads. The naked area of electrodes exposed to the liquid is defined by the opening in the solder mask. Since the Moku:Go instrument comes with crocodile clips for the power supply and two oscilloscope probes for signals, simple square pads along the edges of the dry regions are used for connections (avoiding expensive connectors). For coupling with the 3D-printed fluidics, placed on top of the PCB, a rubber O-ring of 28 mm diameter was chosen and four additional M3 screws are used to tighten the sandwich.

Since the opamps had to be purchased in advance, also given the unpredictable delivery issues due to the silicon

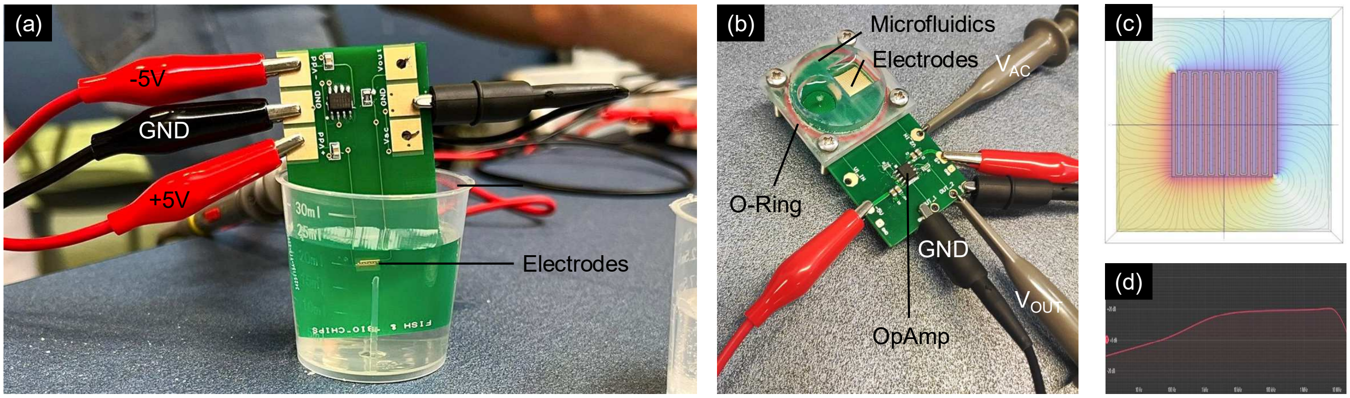

Fig. 6. Example of the PCB characterization with the beaker (a, infinite water thickness) and with a microfluidic chamber (b, finite thickness). The detailed electric field profile of interdigitated electrodes was studied with COMSOL simualtions (c). All the teams were able to measure the admittance spectrum (d).

shortage, we decided to fix the opamp model, common to all projects: AD8022 by Analog Devices. It has good performance (130 MHz GBWP, 2.5 nV/sqrt(Hz) input equivalent voltage noise, 1.5 mV offset, suitable for this application) and comes in dual package, allowing to use the second opamp freely (for instance to create a safety buffer or a second TIA channel).

D. *Software*

Several examples of free software for PCB design exist (from Eagle to the most recent KiCad and CircuitMaker). Among the most professional ones, Altium Designer® is now freely available for students. For this activity, given the ease to merge several 2-layer layouts, we selected DesignSpark, freely available from the website of RS Components.

IV. Laboratory Execution

A total of 14 teams took part in the activity, mentored by the teacher and a teaching assistant. Figure 6 shows two examples of the boards realized by the students during the characterization in the lab (a,b). Deionized and tap water were used as starting solution and then salt was added to track the value of R_{ION} with impedance measurements. The solution conductivity is cross-checked with a pen-type low-cost conductivity meter. Some groups have proposed to measure the change of conductivity with temperature (~2%/°C) and, thus, a heating plate was also used to warm the solution.

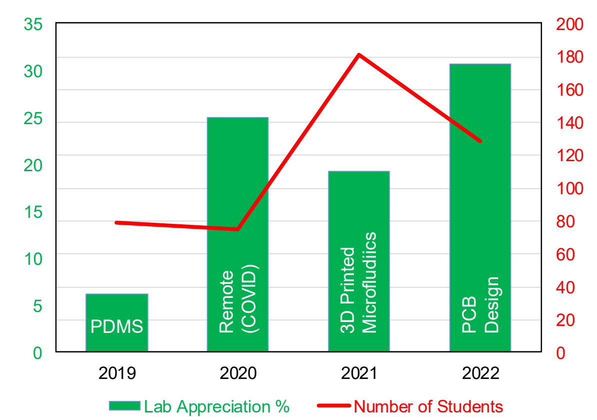

Fig. 7. Growth of the number of students attending the course (red line) and of the percentage of students positively evaluating the laboratory (gree bars).

V. Assesment and Discussion

In order to assess the impact of this new laboratory activity in a quantitative way, we consider the number of explicit and positive mentions to the laboratory activity in the comments section of the anonymous satisfaction surveys mandatorily filled by the students at the end of the course. This figure of merit is normalized on the total number of students (red line), which has been growing in the last years. As reported in Fig. 7, in 2022, after the first edition of the proposed activity, the appreciation (green bars) has exceeded 30% with respect to an average of ~16% of the previous years. Previous editions of the laboratory focused on microfluidics: the teams had to design microchannels (such as fluidic mixers) and then characterized them in the lab by means of syringe pumps, colored fluids and USB microscopes. Up to 2019 the fabrication material was PDMS (polydimethylsiloxane), the standard in the field of microfluidics. Despite several key advantages for research use (optical transparence, biocompatibility) this material presents some limitations from the educational point of view among which the need for a master, for plasma bonding and cost. In 2021, we moved to stereolithographic 3D printing, which is more versatile despite not reaching sub-200 µm resolution.

The total materials cost of this 2-month experience is composed of non-recurring costs: the Moku:Go instrument ($600 per team and comparable to other platforms such as Red Pitaya) and the conductivity meter ($100), and by recurring costs: PCB manufacturing ($200), electronics components ($20 per team). Thus, considering 10 teams of 5 persons (for a total of ~50 students) the initial investment is of $6500, while the successive iterations of the lab cost ~$400 per year.

VI. Conclusion

A multidisciplinary laboratory experience merging basic PCB design, circuit simulation, conformal mapping, fluidic design, and impedance measurements in liquid has been proposed. It was positively evaluated by students (facing PCB design for the first time). The design activity was carried out at home by teams for about 1 month. The classroom time was 10 hours. This time should be increased to cover additional aspects (such as tutorials of the software tools), possibly leveraging on-line post-pandemic learning formats [20].

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